

## SAFETY DATA SHEET

Creation Date 04-June-2010

Revision Date 24-December-2021

Revision Number 6

### 1. Identification

**Product Name** Boron atomic absorption standard solution, 1 mg/ml B in water

**Cat No. :** AC195860000; AC195861000; AC195865000

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Not classified under the Hazardous Products Regulations (SOR/2015-17)

Not a dangerous substance or mixture according to the  
Globally Harmonized System (GHS)

#### Label Elements

None required

### 3. Composition/Information on Ingredients

| Component          | CAS-No     | Weight % |
|--------------------|------------|----------|
| Water              | 7732-18-5  | 99-99.5  |
| Boric acid (H3BO3) | 10043-35-3 | 0.5-1    |

#### 4. First-aid measures

|                                                               |                                                                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                                         | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                                             | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                                              | Clean mouth with water and drink afterwards plenty of water.                                                      |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | None reasonably foreseeable.<br>Treat symptomatically                                                             |

#### 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Alcohol resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | No information available                                                              |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | No information available                                                              |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

#### **Specific Hazards Arising from the Chemical**

None reasonably foreseeable.

#### **Hazardous Combustion Products**

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### **NFPA**

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 0             | 0                   | 0                  | N/A                     |

#### 6. Accidental release measures

|                                  |                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required.                        |
| <b>Environmental Precautions</b> | Should not be released into the environment. See Section 12 for additional Ecological Information. |

**Methods for Containment and Clean Up** Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

## 7. Handling and storage

|                 |                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Store indoors. Incompatible Materials. Strong oxidizing agents.                 |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                                    | Alberta | British Columbia                                      | Ontario TWAEV                                         | Quebec                                                | ACGIH TLV                                             | OSHA PEL | NIOSH IDLH |
|----------------------------------------------|---------|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|----------|------------|
| Boric acid (H <sub>3</sub> BO <sub>3</sub> ) |         | TWA: 2 mg/m <sup>3</sup><br>STEL: 6 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup><br>STEL: 6 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup><br>STEL: 6 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup><br>STEL: 6 mg/m <sup>3</sup> |          |            |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

### Engineering Measures

None under normal use conditions.

### Personal protective equipment

|                        |                                                    |
|------------------------|----------------------------------------------------|
| <b>Eye Protection</b>  | Wear safety glasses with side shields (or goggles) |
| <b>Hand Protection</b> | Protective gloves                                  |

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                       |                          |
|-----------------------|--------------------------|
| <b>Physical State</b> | Liquid                   |
| <b>Appearance</b>     | No information available |
| <b>Odor</b>           | Odorless                 |

|                                        |                            |
|----------------------------------------|----------------------------|
| Odor Threshold                         | No information available   |
| pH                                     | No information available   |
| Melting Point/Range                    | 0 °C / 32 °F               |
| Boiling Point/Range                    | 100 °C / 212 °F @ 760 mmHg |
| Flash Point                            | No information available   |
| Evaporation Rate                       | No information available   |
| Flammability (solid,gas)               | Not applicable             |
| Flammability or explosive limits       |                            |
| Upper                                  | No data available          |
| Lower                                  | No data available          |
| Vapor Pressure                         | No information available   |
| Vapor Density                          | > 1.0                      |
| Specific Gravity                       | 1                          |
| Solubility                             | Soluble in water           |
| Partition coefficient; n-octanol/water | No data available          |
| Autoignition Temperature               | No information available   |
| Decomposition Temperature              | No information available   |
| Viscosity                              | No information available   |

## 10. Stability and reactivity

|                                  |                                                                          |
|----------------------------------|--------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                               |
| Stability                        | Stable under normal conditions.                                          |
| Conditions to Avoid              | Incompatible products.                                                   |
| Incompatible Materials           | Strong oxidizing agents                                                  |
| Hazardous Decomposition Products | Thermal decomposition can lead to release of irritating gases and vapors |
| Hazardous Polymerization         | No information available.                                                |
| Hazardous Reactions              | None under normal processing.                                            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Oral LD50   | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Dermal LD50 | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Vapor LC50  | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |

#### Component Information

| Component          | LD50 Oral          | LD50 Dermal             | LC50 Inhalation |
|--------------------|--------------------|-------------------------|-----------------|
| Water              | -                  | -                       | -               |
| Boric acid (H3BO3) | 2660 mg/kg ( Rat ) | > 2000 mg/kg ( Rabbit ) | Not listed      |

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------|-----------|------------|------------|------------|------------|------------|
| Water     | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                    |            |            |            |            |            |            |
|--------------------|------------|------------|------------|------------|------------|------------|
| Boric acid (H3BO3) | 10043-35-3 | Not listed | Not listed | Not listed | Not listed | Not listed |
|--------------------|------------|------------|------------|------------|------------|------------|

**Mutagenic Effects** No information available

**Reproductive Effects** Contains ingredients that are suspected reproductive hazards.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

| Component          | Freshwater Algae | Freshwater Fish                          | Microtox | Water Flea                                   |
|--------------------|------------------|------------------------------------------|----------|----------------------------------------------|
| Boric acid (H3BO3) | -                | Gambusia affinis: LC50:<br>5600 mg/L/96h | -        | EC50: 115 - 153 mg/L, 48h<br>(Daphnia magna) |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component          | log Pow |
|--------------------|---------|
| Boric acid (H3BO3) | -0.757  |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

### International Inventories

| Component          | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory<br>notification -<br>Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------|------------|-----|------|------|-----------------------------------------------------|-----------|--------|-----|
| Water              | 7732-18-5  | X   | -    | X    | ACTIVE                                              | 231-791-2 | -      | -   |
| Boric acid (H3BO3) | 10043-35-3 | X   | -    | X    | ACTIVE                                              | 233-139-2 | -      | -   |

| Component                                    | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Water                                        | 7732-18-5  | X     | KE-35400 | X    | -    | X    | X    | X     | X     |
| Boric acid (H <sub>3</sub> BO <sub>3</sub> ) | 10043-35-3 | X     | KE-03499 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations**

**Authorisation/Restrictions according to EU REACH**

| Component                                    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Boric acid (H <sub>3</sub> BO <sub>3</sub> ) | -                                                                   | Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - 233-139-2 - Toxic for reproduction, Article 57c                                 |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                    | CAS-No     | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------------------|------------|----------|------------------------------|---------------------------|--------------------------------------------|
| Water                                        | 7732-18-5  | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Boric acid (H <sub>3</sub> BO <sub>3</sub> ) | 10043-35-3 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component                                    | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water                                        | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Boric acid (H <sub>3</sub> BO <sub>3</sub> ) | 10043-35-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

**Prepared By**

Regulatory Affairs

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|                         |                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Creation Date</b>    | 04-June-2010                                                                                                                                                                       |
| <b>Revision Date</b>    | 24-December-2021                                                                                                                                                                   |
| <b>Print Date</b>       | 24-December-2021                                                                                                                                                                   |
| <b>Revision Summary</b> | This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals. |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**